Project Title: J.A.R.V.I.S.: Intelligent Multi-Modal Desktop AI Assistant with Computer Vision Gesture Telemetry, Real-Time Voice Synthesis, and Autonomous System Automation

Project Batch No.: 21CS15

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Abstract

Modern desktop computing environments often require manual keyboard and mouse interactions that can hinder seamless multitasking and touchless control. This project aims to design and implement J.A.R.V.I.S., an intelligent multi-modal desktop assistant and personal user interface (PUI) integrating conversational artificial intelligence, optical computer vision, and operating-system-level telemetry automation. The system leverages state-of-the-art Large Language Models (LLMs) via OpenRouter API for reasoning, contextual query resolution, and multi-turn dialogues, combined with real-time speech recognition and text-to-speech (TTS) audio synthesis. An integrated Computer Vision engine built on OpenCV and optical telemetry enables touchless hand gesture controls including master mute freeze, HUD scaling, rotary volume dial, window switching, stealth lock, and air-mouse reticle navigation. On the backend, a Python-based asynchronous telemetry server interfacing with Windows APIs (via ctypes, winreg, psutil, pycaw, and screen_brightness_control) enables direct system management including hardware monitoring, volume and brightness control, automated presentation and PDF generation (using reportlab and python-pptx), and zero-latency application indexing. The frontend delivers a futuristic, cybernetic webview HUD with responsive visual feedback, sound cues, and interactive widgets. The expected outcome is a unified, low-latency multi-modal AI assistant that enhances user productivity, accessibility, and human-computer interaction across academic, creative, and professional workflows.

Problem Statement

Traditional computer interaction models depend heavily on physical input devices, creating friction in hands-free environments, multitasking scenarios, and accessible computing. Furthermore, conventional voice assistants lack deep operating-system telemetry integration, real-time computer vision gesture control, and autonomous document generation capabilities within a unified desktop environment.

Keywords

Multi-Modal AI, Computer Vision, Hand Gesture Recognition, Desktop Automation, Large Language Models (LLM), Speech Recognition, OpenCV, Human-Computer Interaction (HCI)

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